# Configure ASA Basic Settings and Firewall Using the CLI

# Report by Ezekiel Waweru CNS06-24049

## INTRODUCTION

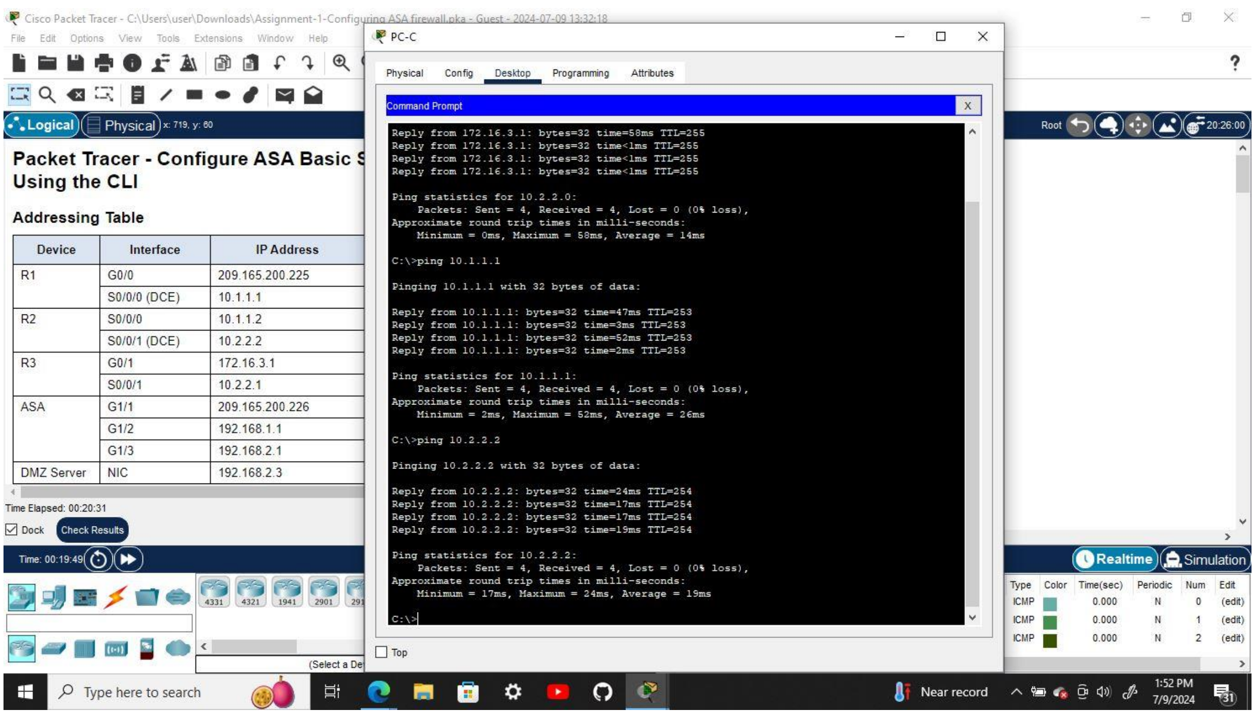
This was an assignment on configuring basic settings and a firewall on a Cisco Adaptive Security Appliance (ASA) using the Command Line Interface (CLI) by using a set of commands that set up a devices essential and network security functions.

## Methodology

### Part 1: Verify Connectivity and Explore the ASA

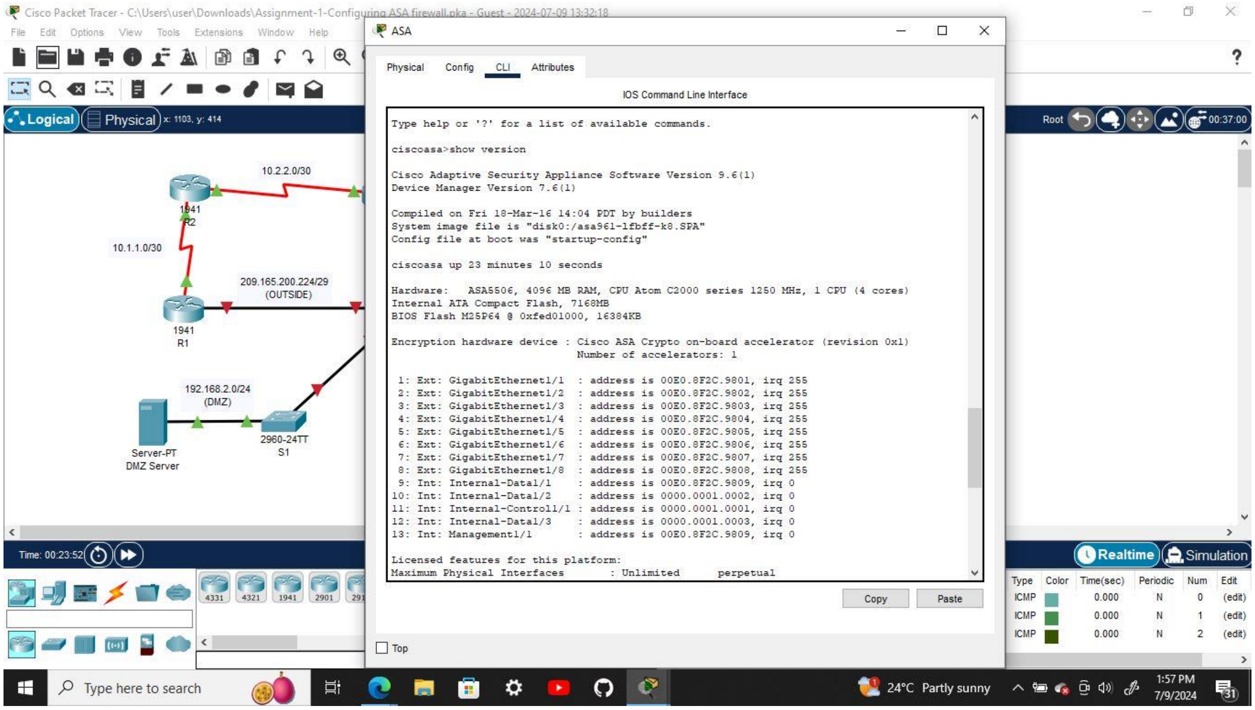
### Step 1: Verify connectivity.

The ASA is not currently configured. However, all routers, PCs, and the DMZ server are configured. Verify that PC-C can ping any router interface. PC-C is unable to ping the ASA, PC-B, or the DMZ server.




### Step 2: Determine the ASA version, interfaces, and license.

Use the **show version** command to determine various aspects of this ASA device.




### Step 3: Determine the file system and contents of flash memory.

a. Enter privileged EXEC mode. A password has not been set. Press **Enter** when prompted for a password.

b. Use the **show file system** command to display the ASA file system and determine which prefixes are supported.

c. Use the **show flash:** or **show disk0:** command to display the contents of flash memory.

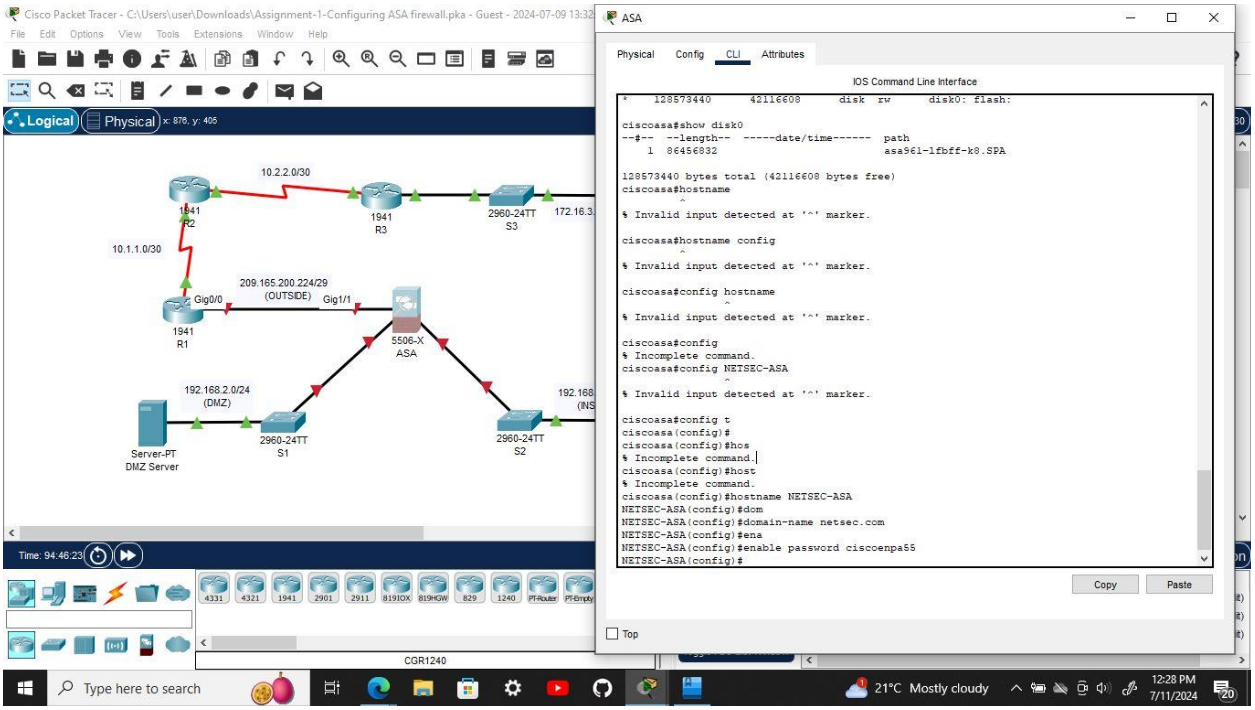
## Part 2: Configure ASA Settings and Interface Security Using the CLI

**Tip**: Many ASA CLI commands are similar to, if not the same, as those used with the Cisco IOS CLI. In addition, the process of moving between configuration modes and submodes is essentially the same.

### Step 1: Configure the hostname and domain name.

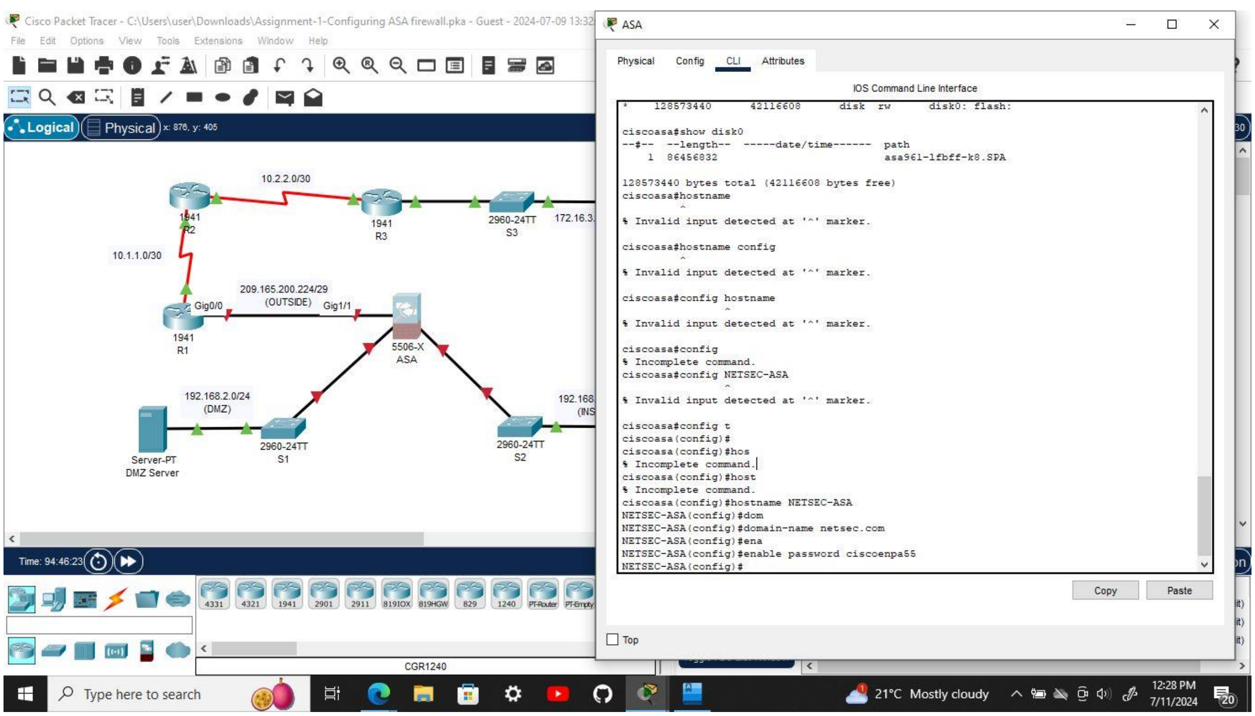
a. Configure the ASA hostname as **NETSEC-ASA**.

b. Configure the domain name as **netsec.com**.



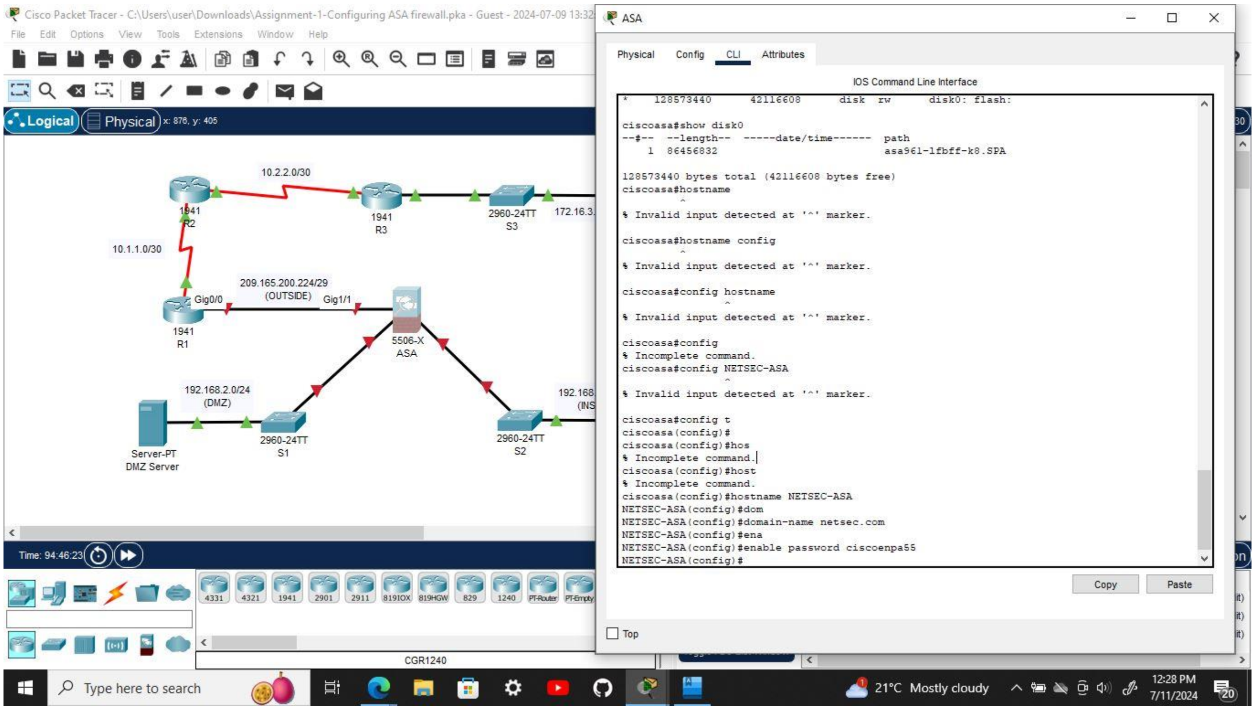

## Step 2: Configure the enable mode password.

Use the **enable password** command to change the privileged EXEC mode password to **ciscoenpa55**.




## Step 3: Set the date and time.

Use the **clock set** command to manually set the date and time (this step is not scored).

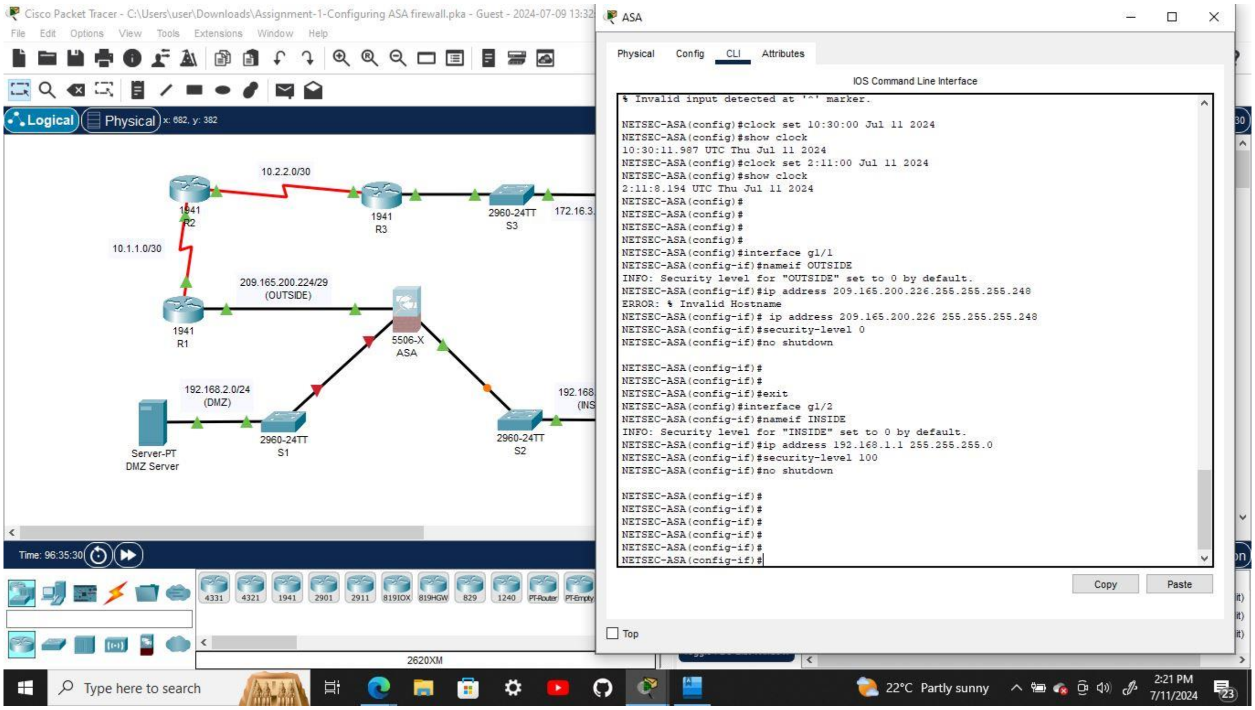



## Step 4: Configure the INSIDE and OUTSIDE interfaces.

You will only configure the G1/1 (OUTSIDE) and G1/2 (INSIDE) interfaces at this time. The G1/3 (DMZ) interface will be configured in Part 5 of the activity.

a. Create the G1/1 interface for the outside network (209.165.200.224/29), set the security level to the lowest setting of 0, and enable the interface.

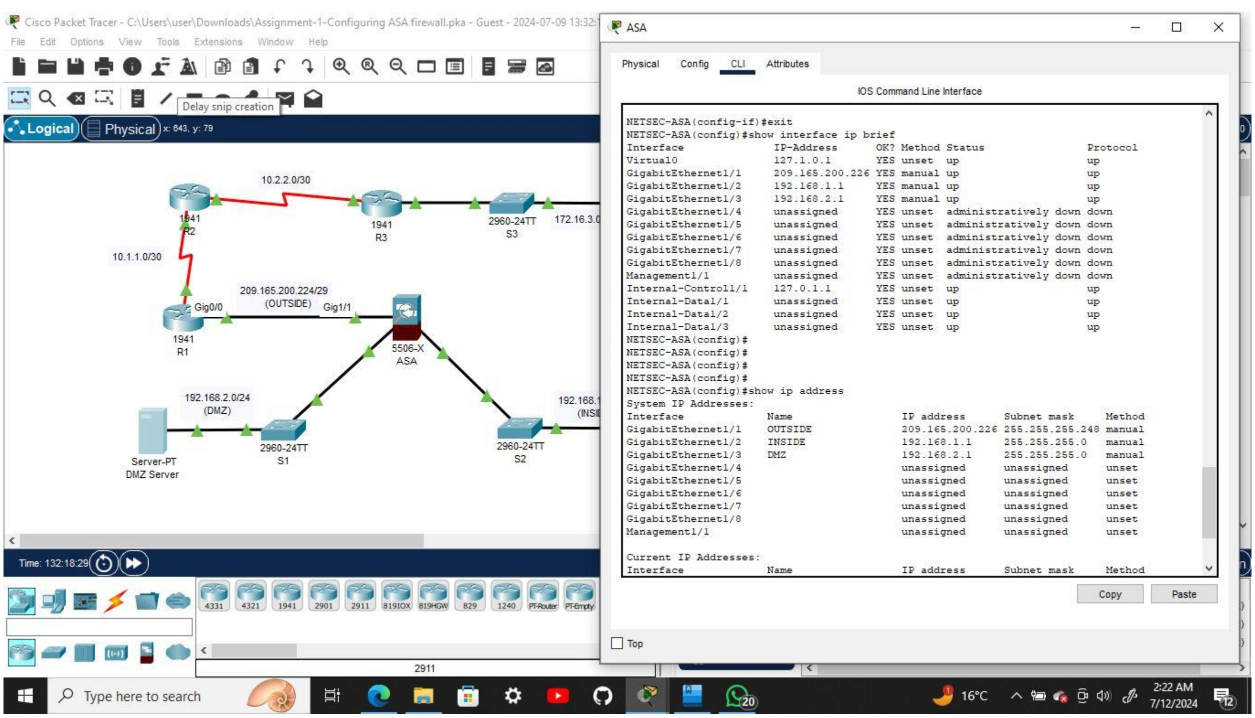
b. Configure the G1/2 interface for the inside network (192.168.1.0/24) and set the security level to the highest setting of 100 and enable the interface.




c. Use the following verification commands to check your configurations:

1) Use the **show interface ip brief** command to display the status for all ASA interfaces.

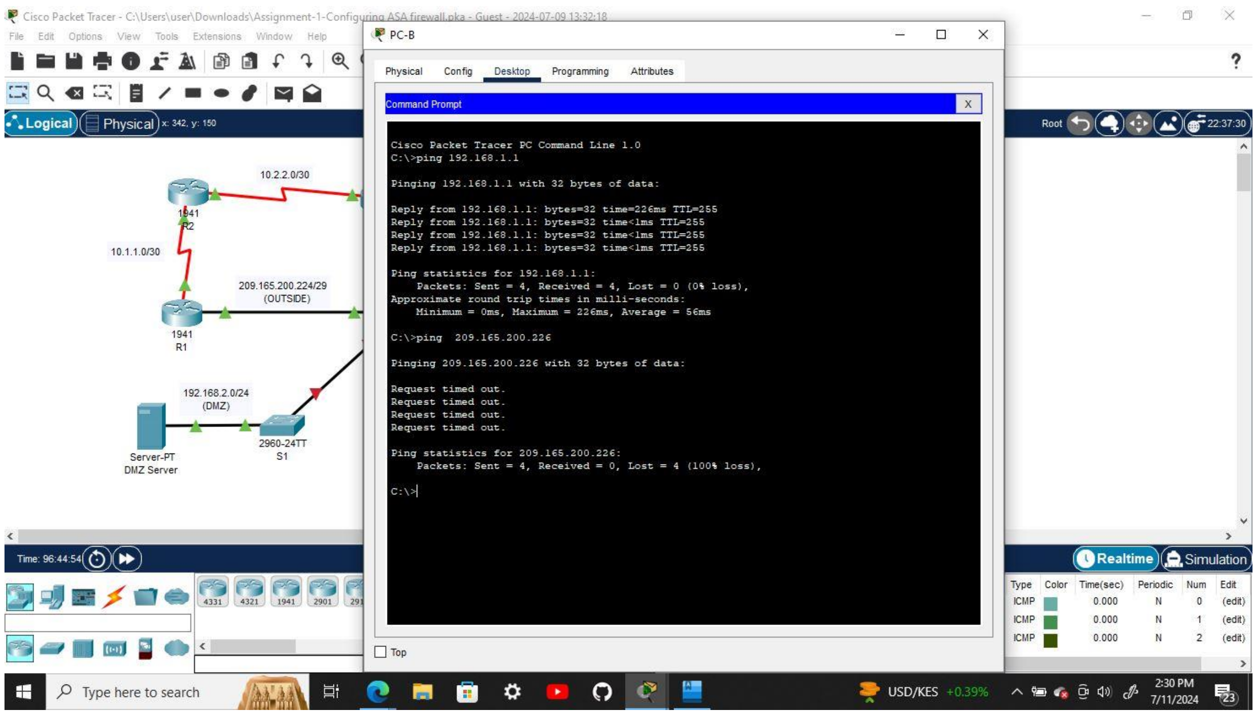
2) Use the **show ip address** command to display the interface information.



**Step 5: Test connectivity to the ASA.**

a. You should be able to ping from PC-B to the ASA inside interface address (192.168.1.1). If the pings fail, troubleshoot the configuration as necessary.

b. From PC-B, ping the G1/1 (OUTSIDE) interface at IP address 209.165.200.226. You should not be able to ping this address.

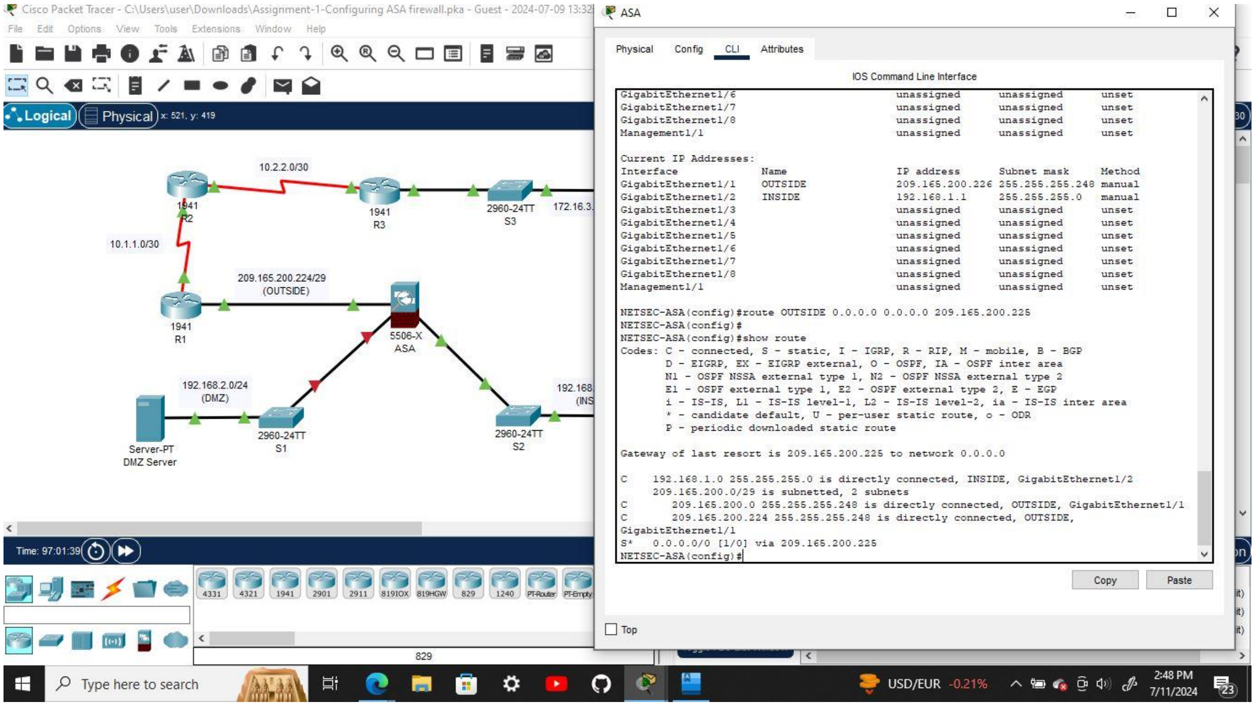



**Part 3: Configure Routing, Address Translation, and Inspection Policy Using the CLI**

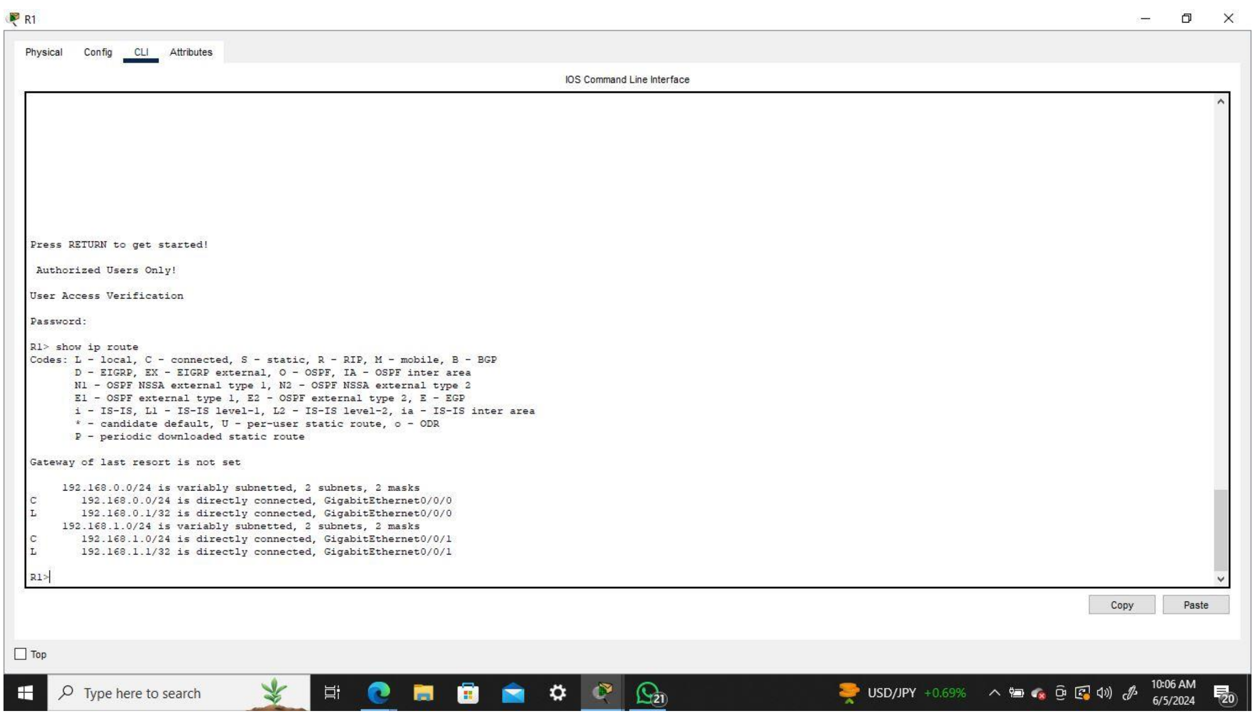
**Step 1: Configure a static default route for the ASA.**

Configure a default static route on the ASA OUTSIDE interface to enable the ASA to reach external networks.

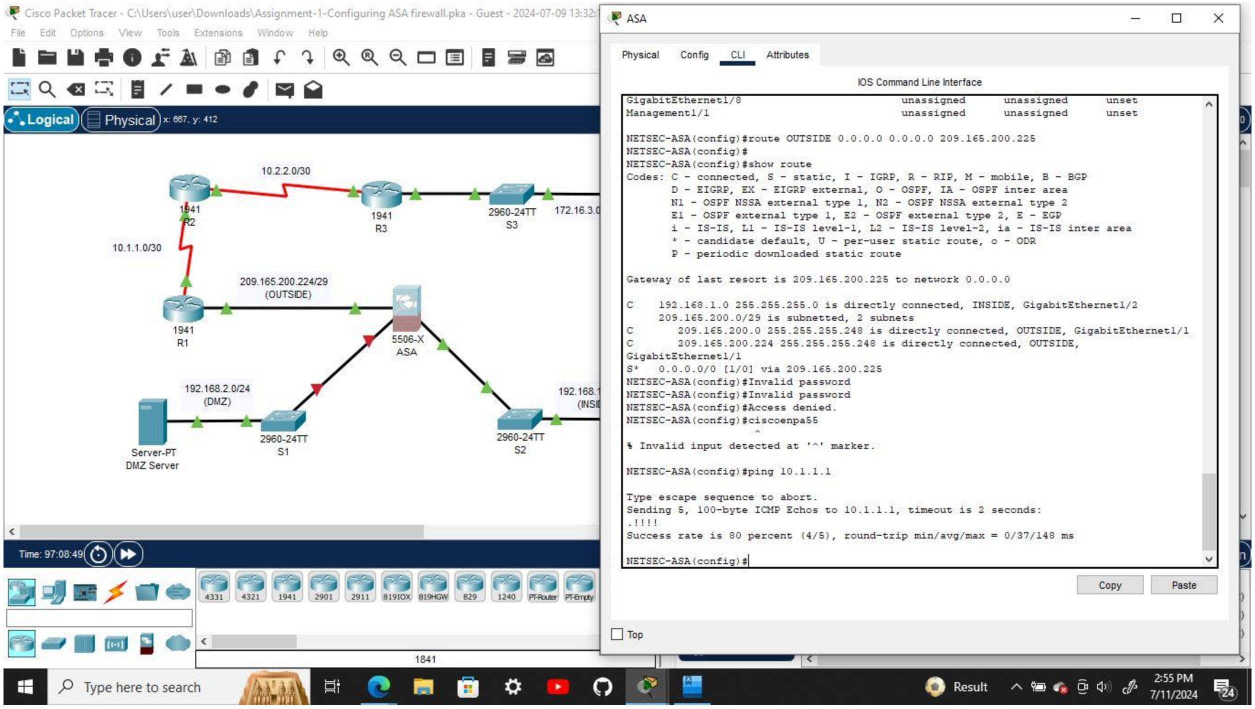
a. Create a “quad zero” default route using the **route** command, associate it with the ASA OUTSIDE interface, and point to the R1 G0/0 IP address (209.165.200.225) as the gateway of last resort.




b. Issue the **show route** command to verify the static default route is in the ASA routing table.



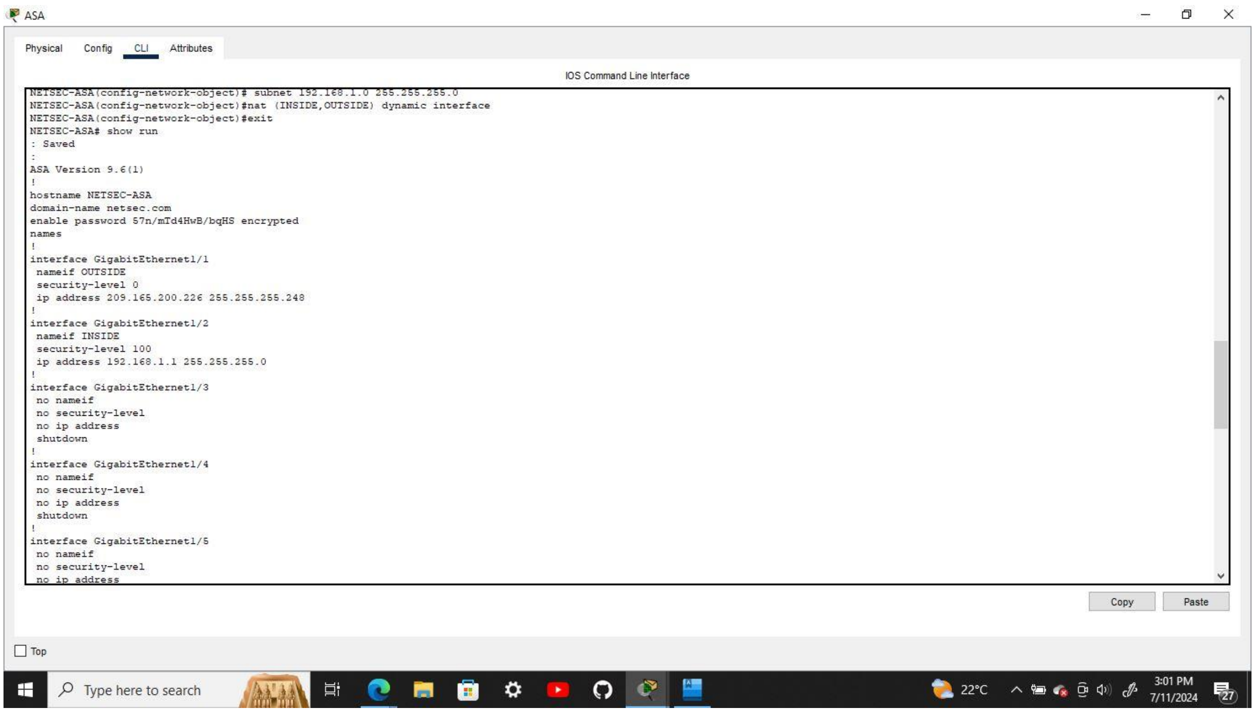

c. Verify that the ASA can ping the R1 S0/0/0 IP address 10.1.1.1. If the ping is unsuccessful, troubleshoot as necessary.



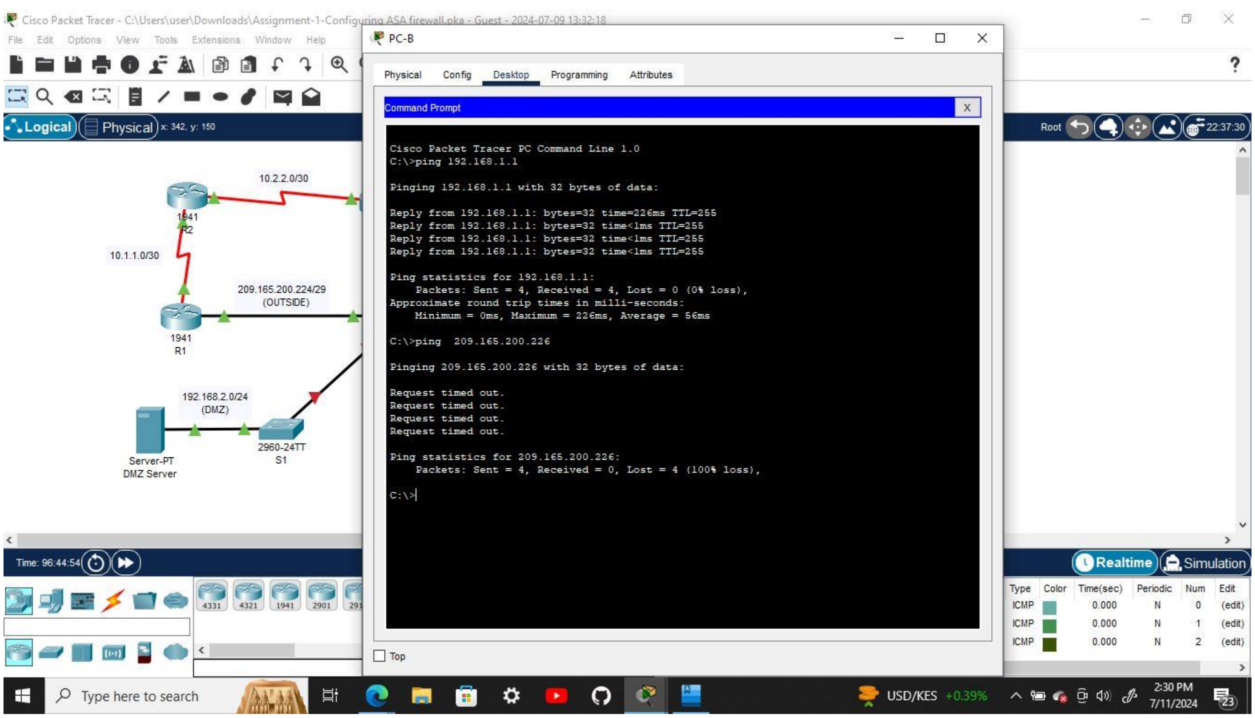

### Step 2: Configure address translation using PAT and network objects.

a. Create network object **INSIDE-NET** and assign attributes to it using the **subnet** and **nat** commands.

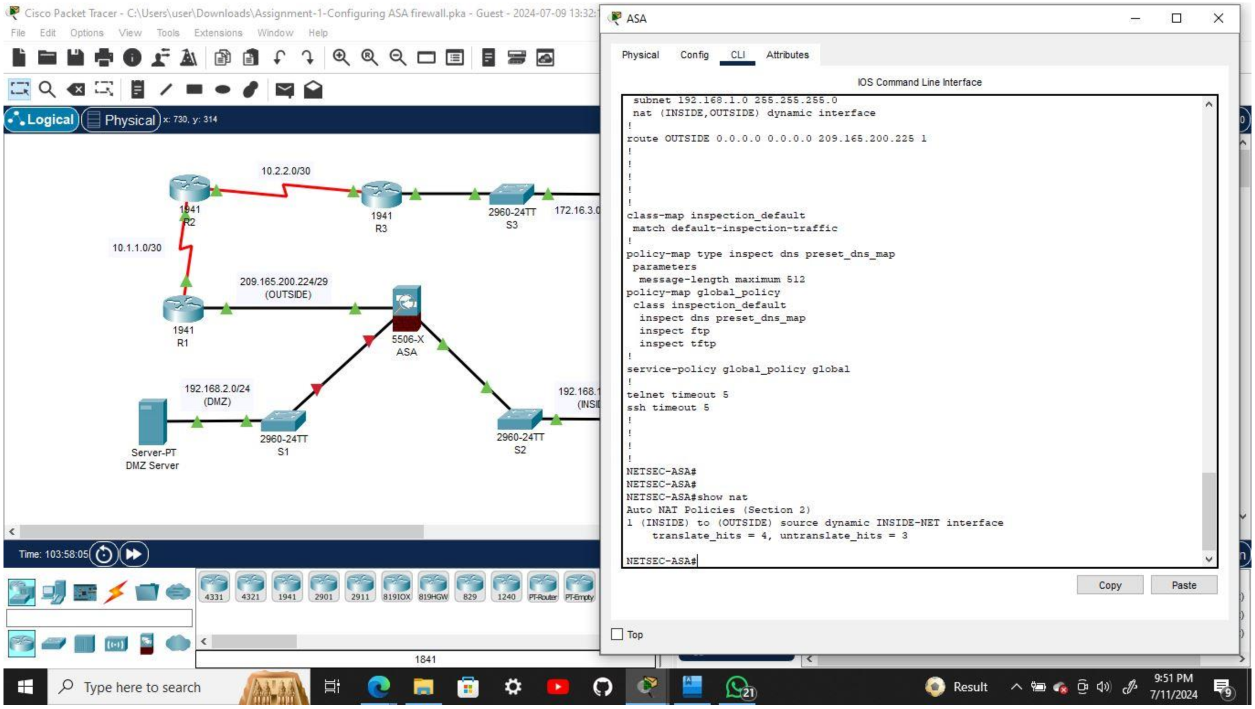
b. The ASA splits the configuration into the object portion that defines the network to be translated and the actual **nat** command parameters. These appear in two different places in the running configuration. Display the NAT object configuration using the **show run** command.




c. From PC-B attempt to ping the R1 G0/0 interface at IP address 209.165.200.225. The pings should fail.




d. Issue the **show nat** command on the ASA to see the translated and untranslated hits. Notice that, of the pings from PC-B, four were translated and four were not. The outgoing pings (echos) were translated and sent to the destination. The returning echo replies were blocked by the firewall policy.



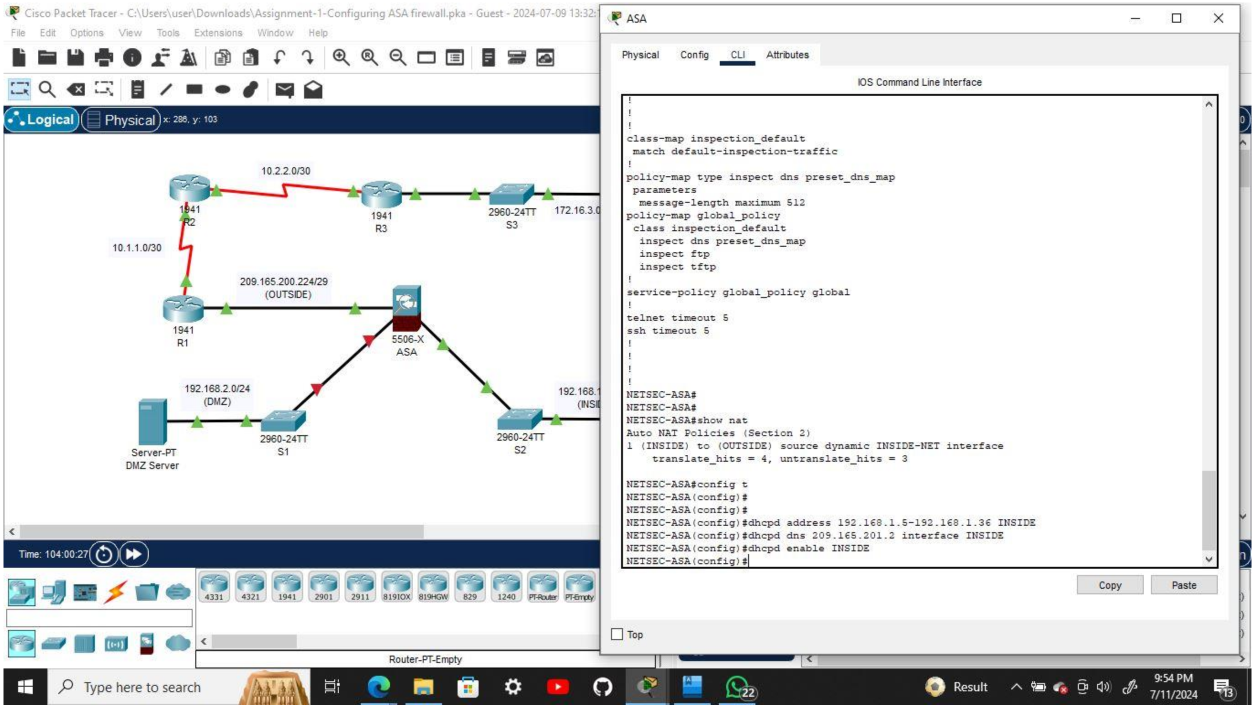

## Part 4: Configure DHCP, AAA, and SSH

### Step 1: Configure the ASA as a DHCP server.

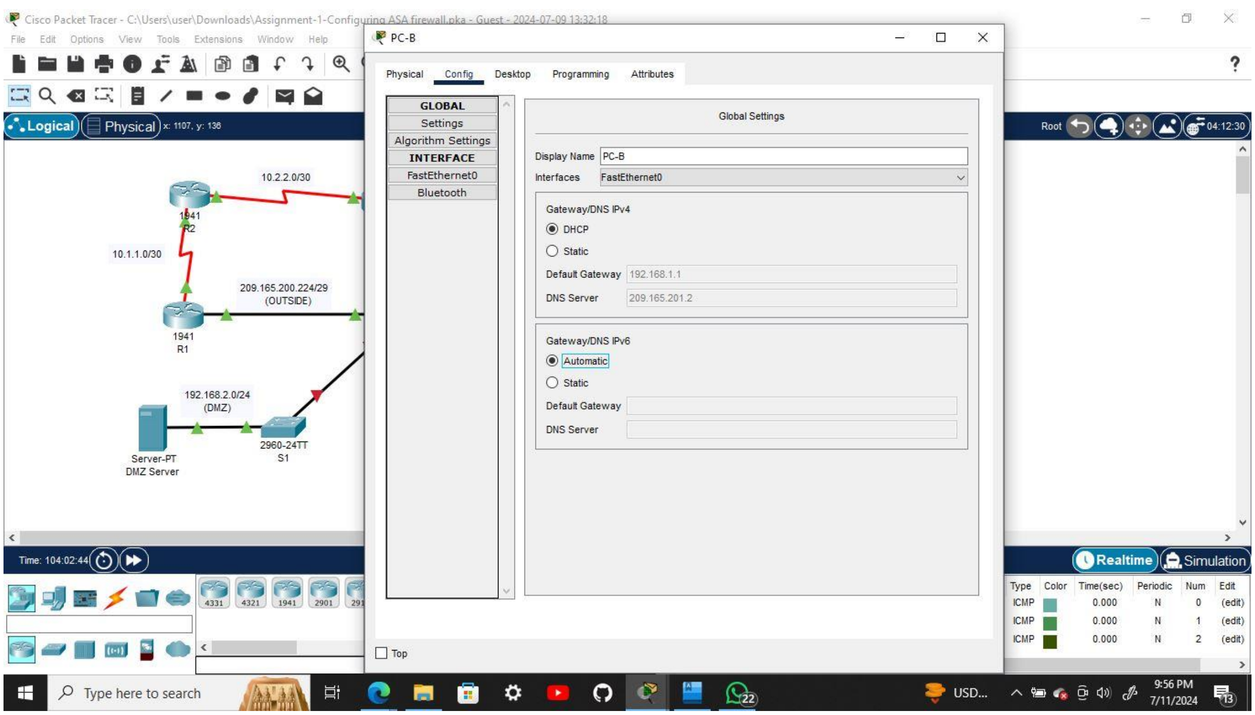
a. Configure a DHCP address pool and enable it on the ASA INSIDE interface.

b. (Optional) Specify the IP address of the DNS server to be given to clients.

c. Enable the DHCP daemon within the ASA to listen for DHCP client requests on the enabled interface (INSIDE).



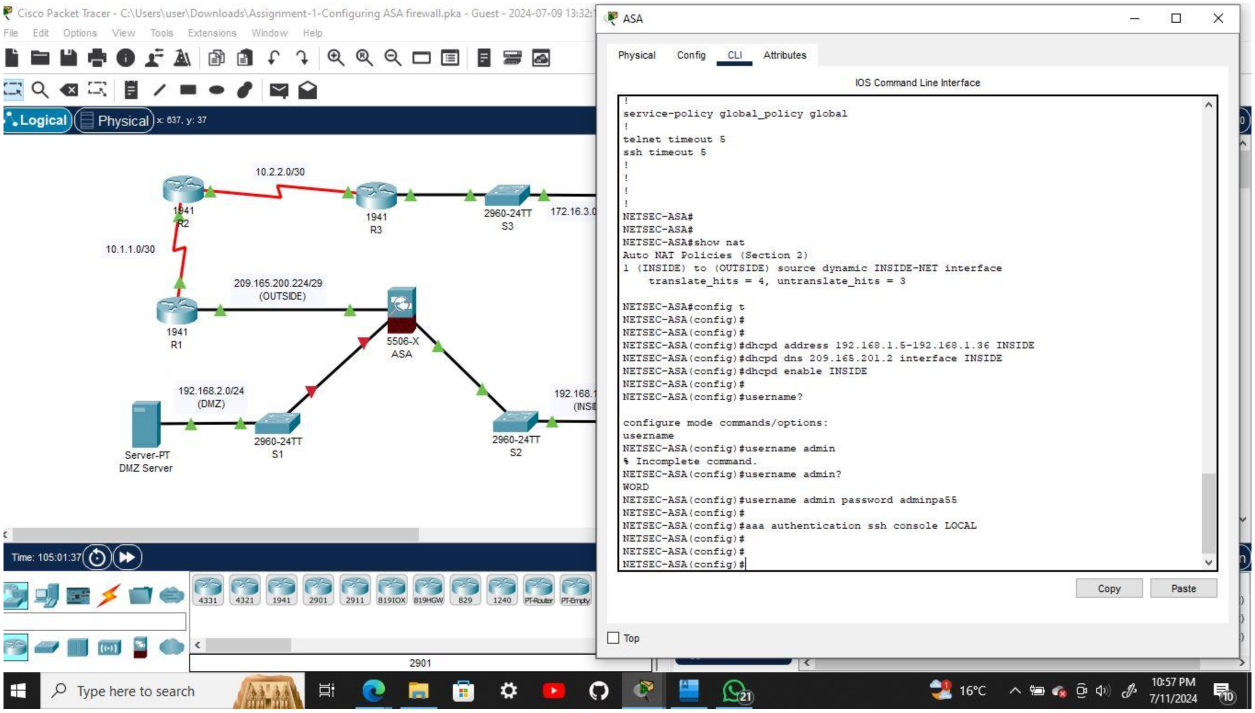

d. Change PC-B from a static IP address to a DHCP client and verify that it receives IP addressing information. Troubleshoot, as necessary to resolve any problems.




**Step 2: Configure AAA to use the local database for authentication.**

a. Define a local user named **admin** by entering the **username** command. Specify a password of **adminpa55**.

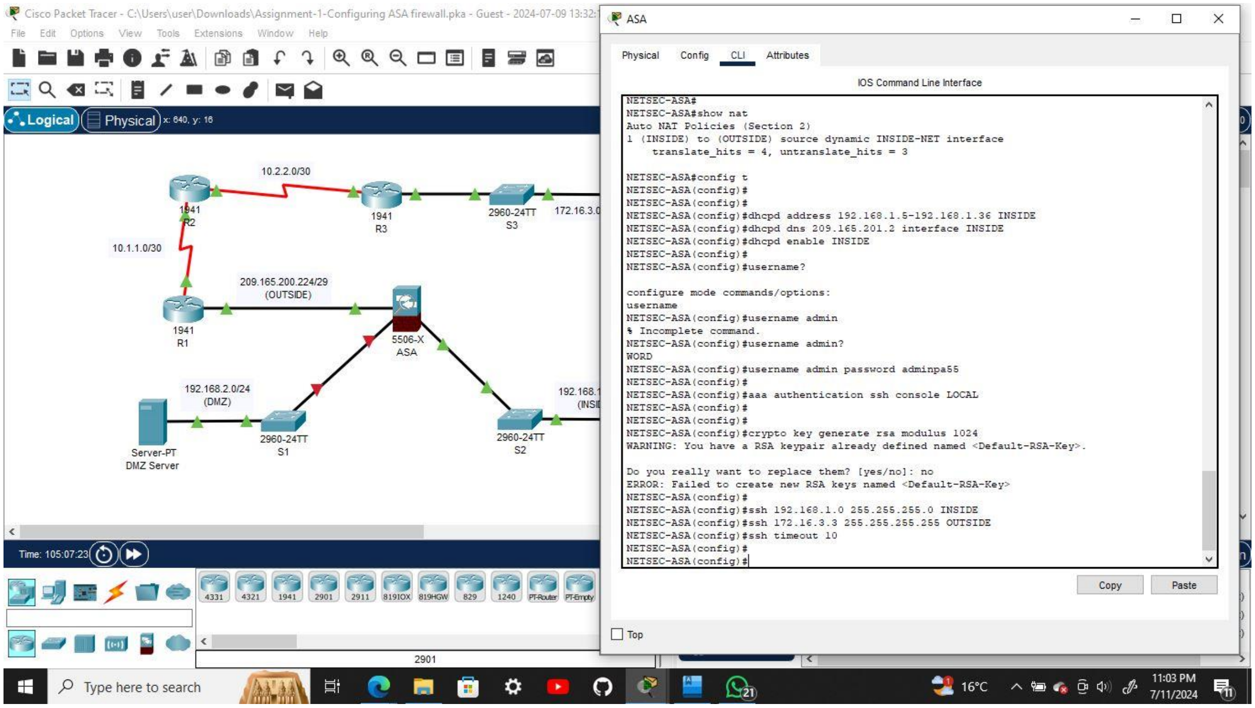
b. Configure AAA to use the local ASA database for SSH user authentication.



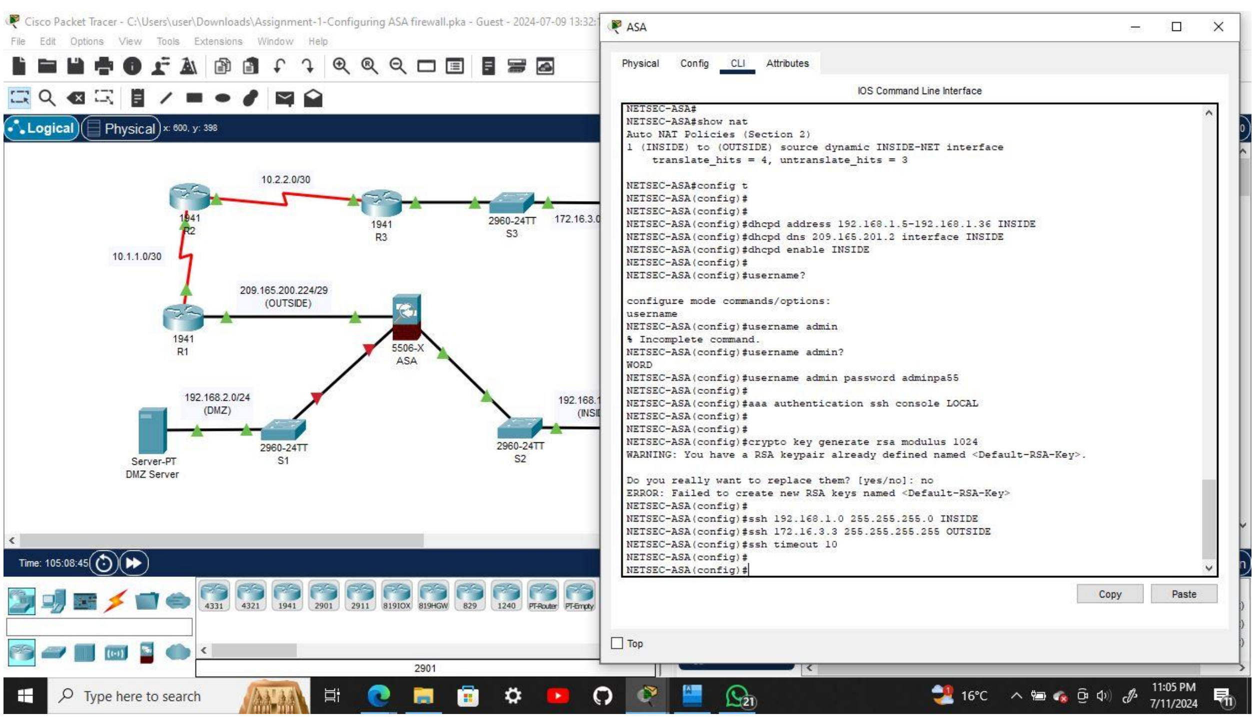

### Step 3: Configure remote access to the ASA.

The ASA can be configured to accept connections from a single host or a range of hosts on the INSIDE or OUTSIDE network. In this step, hosts from the OUTSIDE network can only use SSH to communicate with the ASA. SSH sessions can be used to access the ASA from the inside network.

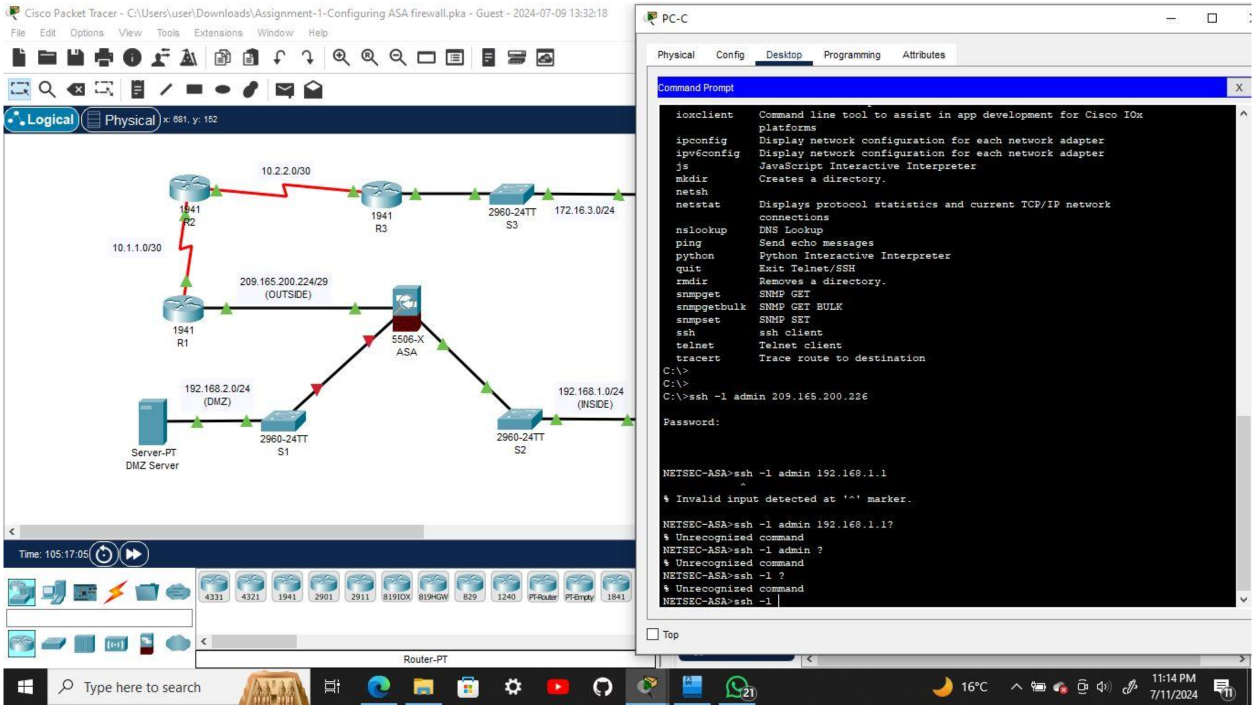
a. Generate an RSA key pair, which is required to support SSH connections. Because the ASA device has RSA keys already in place, enter **no** when prompted to replace them.



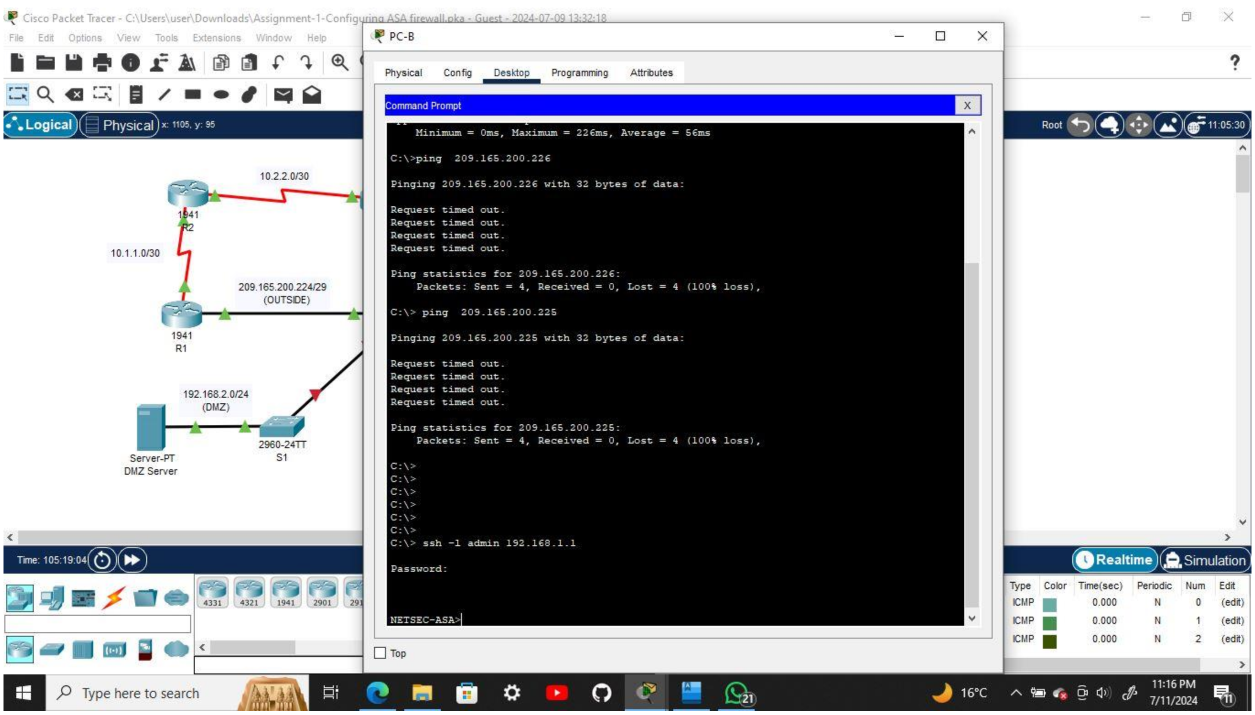
b. Configure the ASA to allow SSH connections from any host on the INSIDE network (192.168.1.0/24) and from the remote management host at the branch office (172.16.3.3) on the OUTSIDE network. Set the SSH timeout to 10 minutes (the default is 5 minutes).



c. Establish an SSH session from PC-C to the ASA (209.165.200.226). Troubleshoot if it is not successful.




d. Establish an SSH session from PC-B to the ASA (192.168.1.1). Troubleshoot if it is not successful.

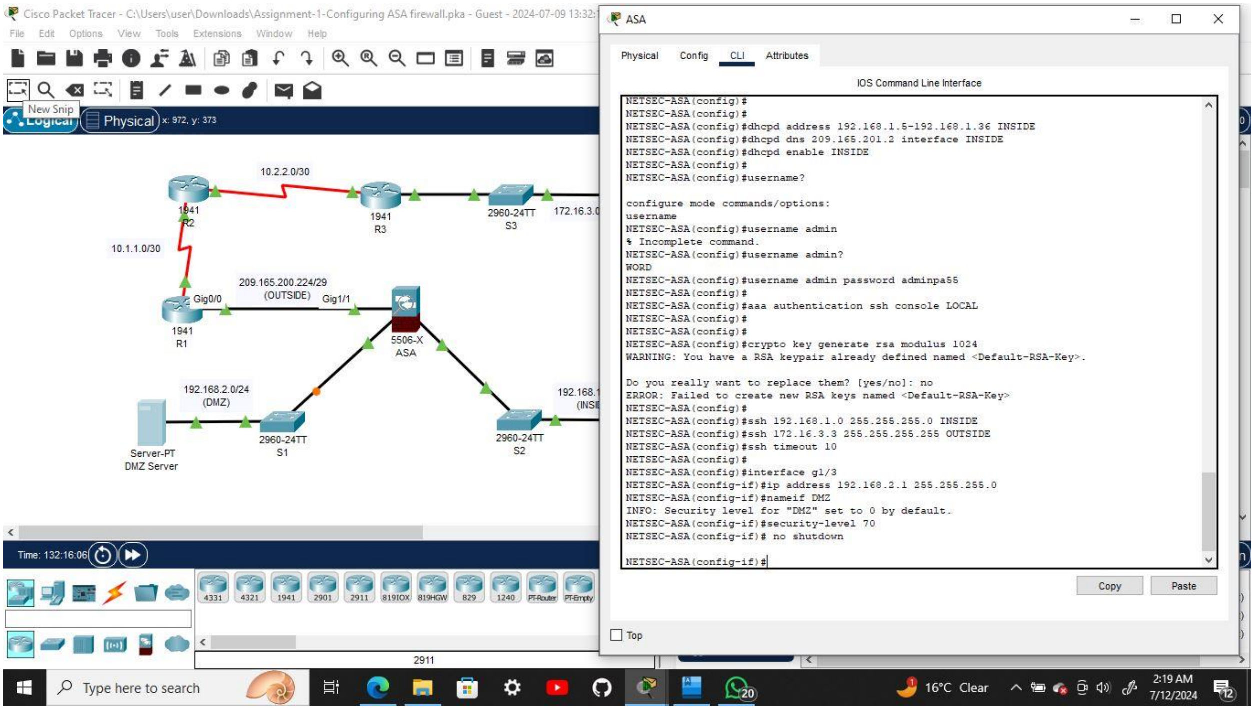



## Part 5: Configure a DMZ, Static NAT, and ACLs

R1 G0/0 and the ASA OUTSIDE interface already use 209.165.200.225 and .226, respectively. You will use public address 209.165.200.227 and static NAT to provide address translation access to the server.

### Step 1: Configure the DMZ interface VLAN 3 on the ASA.

a. Configure DMZ VLAN 3, which is where the public access web server will reside. Assign it IP address 192.168.2.1/24, name it **DMZ**, and assign it a security level of 70. Because the server does not need to initiate communication with the inside users, disable forwarding to interface VLAN 1.




b. Use the following verification commands to check your configurations:

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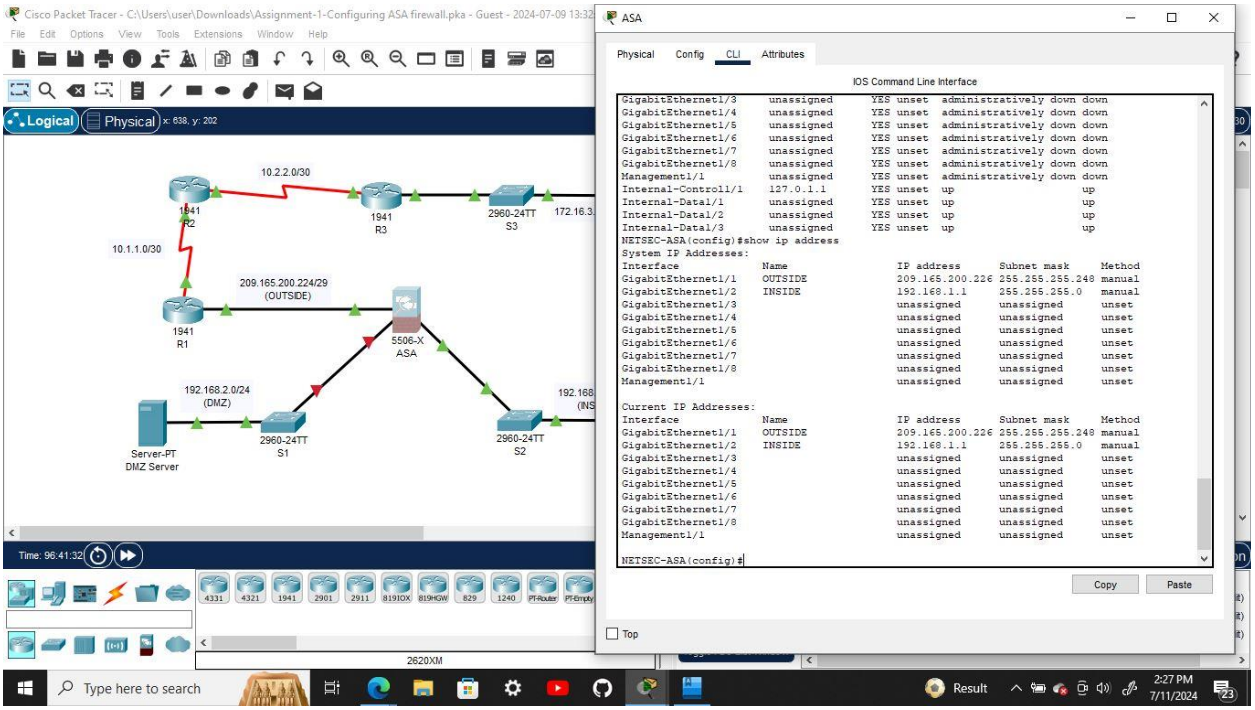
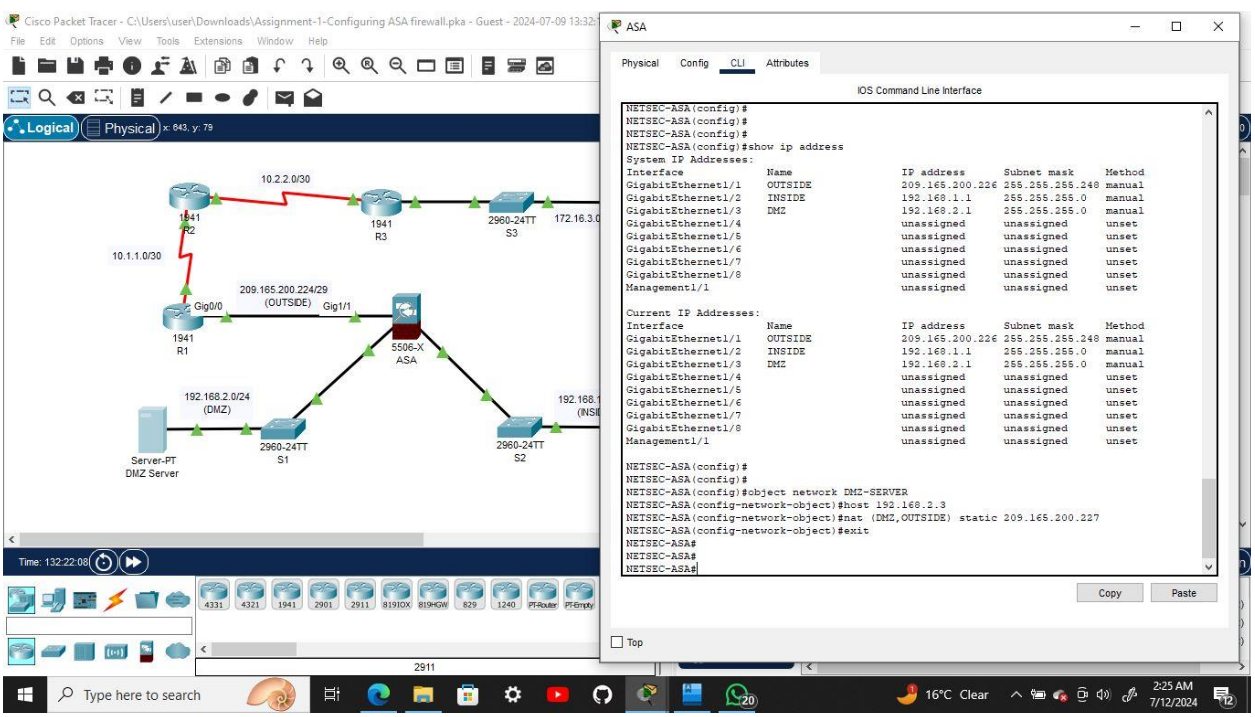


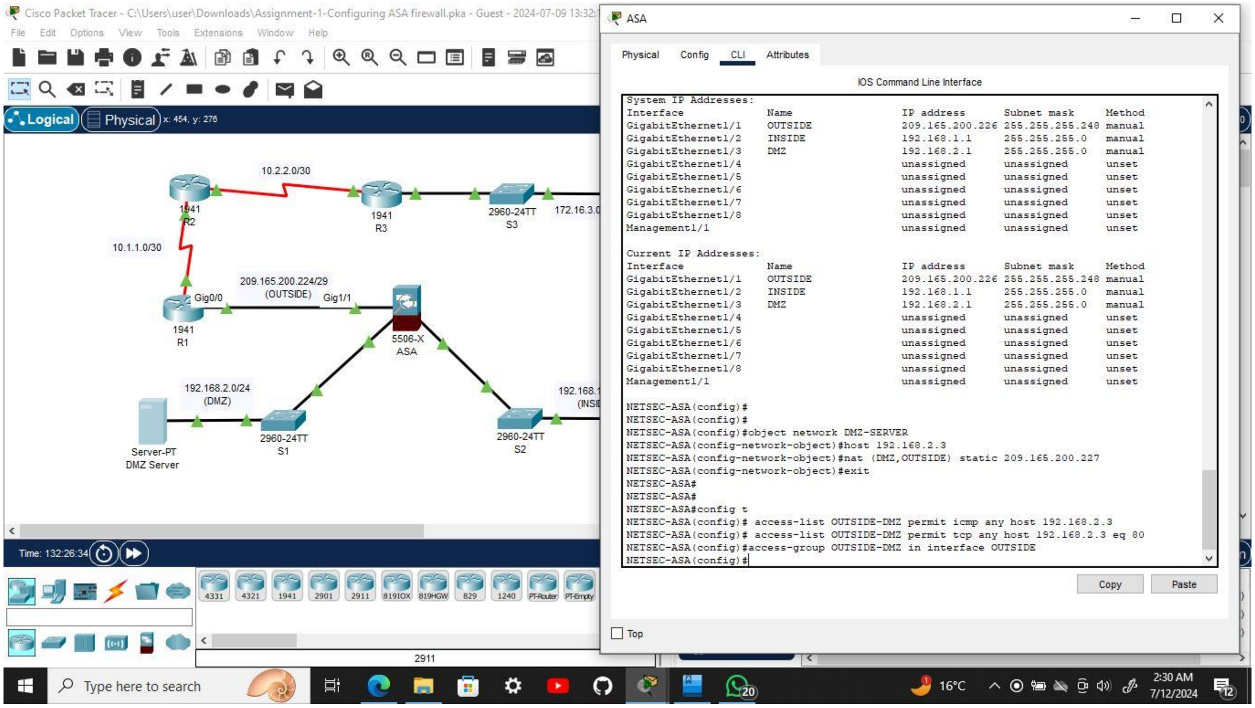

## Step 2: Configure static NAT to the DMZ server using a network object.

Configure a network object named **DMZ-SERVER** and assign it the static IP address of the DMZ server (192.168.2.3). While in object definition mode, use the **nat** command to specify that this object is used to translate a DMZ address to an OUTSIDE address using static NAT, and specify a public translated address of 209.165.200.227.



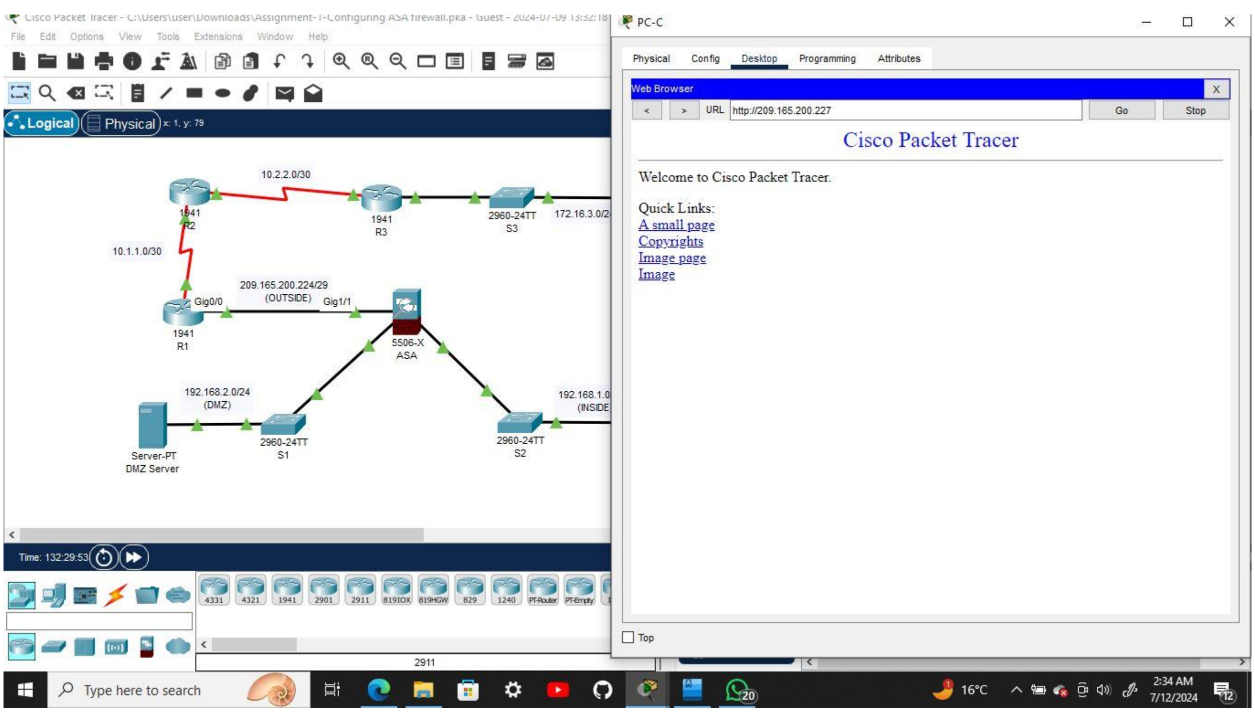
**Step 3: Configure an ACL to allow access to the DMZ server from the Internet.**

Configure a named access list **OUTSIDE-DMZ** that permits the TCP protocol on port 80 from any external host to the internal IP address of the DMZ server. Apply the access list to the ASA OUTSIDE interface in the “IN” direction.




**Step 4: Test access to the DMZ server.**

From a web browser on PC-C, navigate to the DMZ server (209.165.200.227). Troubleshoot if it is not successful.



## Conclusion

This assignment was a learning experience. I got to know more about device configuration. The trouble I ran into while setting the clock also served to humble me as I lingered on it for some minutes. To make sure I got it right after the fifth frustrating try, I looked up how to go about it, and I was able to set the clock and date. It was something elementary but now I'm confident about my ability and will be sure to approach any assignment as someone who knows nothing and id there to learn. I encountered no further trouble and I was able to configure the devices, set up security and as a final step, checked to see the connectivity of all devices. They were all connected and running as expected to my relief.